

Strata PreDeposit Audit Summary

Prepared by Cyfrin

Version 2.1

Lead Auditors

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Contents

1 About Cyfrin 2

2 About This Document 2

3 Disclaimer 2

4 Copyright and License 2

5 Audit Overview 3

6 Audit Scope 3

7 Risk Classification 4

8 Protocol Summary 4

9 Attestation 4

1 About Cyfrin

Cyfrin is a Web3 security company dedicated to bringing industry-leading protection and education to our partners and their projects. Our goal is to create a safe, reliable, and transparent environment for everyone in Web3 and DeFi. Learn more about us at cyfrin.io.

2 About This Document

This document is a public summary of the security audit Cyfrin performed for Strata on Strata PreDeposit. It records the repositories and commits reviewed, the timeline, auditors, in-scope files, and the number and final status of findings by severity. It closes with Cyfrin's attestation of the review's scope and final remediation status.

It is deliberately abbreviated: the technical detail of individual findings is not reproduced here. The complete audit report, version 2.1 dated June 11, 2025, was delivered to Strata, who retain it as the full record of the engagement. This summary was issued on October 29, 2026.

3 Disclaimer

The Cyfrin team makes every effort to find as many vulnerabilities in the code as possible in the given time but holds no responsibility for the findings in this document. A security audit by the team does not endorse the underlying business or product. The audit was time-boxed and the review of the code was solely on the security aspects of the in-scope code being audited. Unless the scope explicitly notes otherwise, the review is limited to the application layer and does not extend to the correctness of API/SDK/execution environment implementations, compilers, standard and third-party libraries, underlying operating systems and hardware.

4 Copyright and License

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5 Audit Overview

Over the course of 3 days, the Cyfrin team conducted an audit on the Strata PreDeposit code provided by Strata. The findings consist of 1 Critical, 1 High, 3 Medium & 16 Low severity issues with the remainder being informational and gas optimizations.

Summary

Project Name	Strata PreDeposit
Repository	contracts
Commit	e053c804f538...
Audit Timeline	May 26th - May 28th, 2025
Methods	Manual Review, Fuzz/Invariant Testing

Issues Found

Severity	Found	Resolved	Acknowledged
Critical Risk	1	1	0
High Risk	1	1	0
Medium Risk	3	3	0
Low Risk	16	16	0
Informational	9	9	0
Gas Optimizations	8	6	2
Total	38	36	2

6 Audit Scope

The scope of this audit is limited to:

```
strata-money-contracts/contracts/predeposit/MetaVault.sol
strata-money-contracts/contracts/predeposit/PreDepositPhaser.sol
strata-money-contracts/contracts/predeposit/PreDepositVault.sol
strata-money-contracts/contracts/predeposit/pUSDeDepositor.sol
strata-money-contracts/contracts/predeposit/pUSDeVault.sol
```

7 Risk Classification

	Impact: High	**Impact: Medium**	**Impact: Low**
Likelihood: High	Critical	High	Medium
Likelihood: Medium	High	Medium	Low
Likelihood: Low	Medium	Low	Low

8 Protocol Summary

Strata is a perpetual yield tranching protocol built on Converge, designed to offer structured yield exposure on USDe, Ethena's delta-neutral synthetic stablecoin. This audit covers the "PreDeposit" functionality, primarily related to these two contracts:

- `pUSDeVault`: a two-phase, multi-asset ERC4626 vault
- `pUSDeDepositor`: a "front-end" or "helper" contract implementing easy deposit options into `pUSDeVault`

The 3 other contracts in the audit scope, `MetaVault`, `PreDepositVault` and `PreDepositPhaser`, are part of the inheritance chain of `pUSDeVault`. The protocol interacts with the following tokens:

- USDe: ERC20, Ethena's synthetic USD token
- sUSDe: ERC4626, yield-bearing equivalent of `USDe`, staked into an ERC4626 vault

pUSDeVault

`pUSDeVault` is a multi-asset ERC4626 vault which can operate in one of two phases:

- Points Phase: accepts and holds `USDe` as the ERC4626 underlying asset, and potentially also other supported assets
- Yield Phase: on activation, redeems all additionally supported assets then stakes the entire `USDe` balance by depositing it into the `sUSDe` ERC4626 vault. Once activated the Yield Phase is permanent, `sUSDe` becomes a supported asset and any future `USDe` deposits are automatically staked in the same way.

pUSDeDepositor

`pUSDeDepositor` is a utility contract which allows users to deposit into `pUSDeVault` using:

- `USDe`: the simplest option, depositing the primary underlying ERC20 asset of `pUSDeVault`
- `sUSDe`: shares from `sUSDe`, which is used as the staking vault for `pUSDeVault`'s `USDe`, only during the yield phase
- `autoSwap to USDe`: other supported stablecoin tokens are swapped for `USDe`, then the swap output `USDe` is deposited into `pUSDeVault` in one transaction
- `supported vaults`: shares from other supported ERC4626 vaults

9 Attestation

Cyfrin attests that it performed a security review of the Strata PreDeposit code provided by Strata.

The repositories, commits, review period and findings with their final status are in the Summary and Issues Found tables in Section 5, and the in-scope files in Section 6. This attestation is limited to the documented scope, audited commits and mitigations specifically reviewed by Cyfrin.

Cyfrin publishes this document and its SHA-256 hash at https://github.com/Cyfrin/cyfrin-audit-reports. To check a copy on macOS or Linux, run the following in the folder that contains it; it prints `OK` when the copy matches. If the copy was renamed, set F to its file name. Before running it, check that the URL is Cyfrin's repository.

```
U=https://github.com/Cyfrin/cyfrin-audit-reports/raw/main/reports/sha256
N=2025-06-11-cyfrin-strata-predeposit-v2.1.pdf
F=$N
H=$(curl -fsSL $U/$N.sha256 | cut -d' ' -f1)
echo "$H *$F" | shasum -a 256 -c
```

Cyfrin's full audit report with all findings provided to Strata has the SHA-256 hash and file name below. To check a copy of it, first check this document as above, then run the following in the folder that contains the full report; it prints `OK` when the copy matches. If the copy was renamed, set F to its file name.

```
H=2839303bbfaa70cc177c387c67e5da573f010d4f6033aa88b2d84ff5cc5cf37c
F=2025-06-11-cyfrin-strata-predeposit-v2.1-full.pdf
echo "$H *$F" | shasum -a 256 -c
```

On Linux, `sha256sum -c` can be used in place of `shasum -a 256 -c`. On Windows, run `Get-FileHash <file> -Algorithm SHA256` in PowerShell and compare the result with the hash.